



System Architecture & System Workflows

Project: ArenaMind AI

Version: 1.0

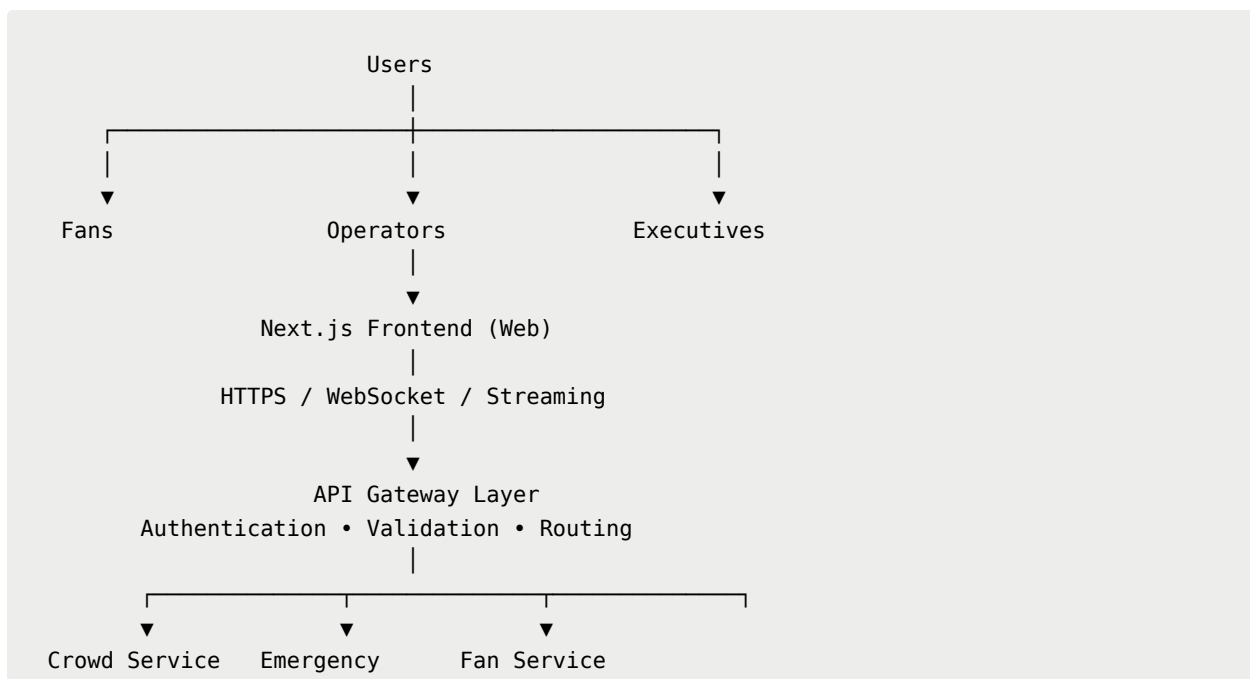
Purpose: High-Level Architecture & Operational Workflows

System Architecture Overview

ArenaMind AI follows an **AI-First, Event-Driven, Cloud-Native Architecture**, where every user interaction and operational event passes through a centralized AI Gateway before reaching domain services.

The platform integrates Generative AI, Digital Twin technology, real-time data, and operational intelligence into one unified system.

High-Level Architecture





Core Modules

Module	Responsibility
AI Gateway	AI orchestration & routing
Digital Twin	Interactive 3D stadium
Crowd Intelligence	Crowd prediction
Fan Assistant	Navigation & recommendations
Emergency Response	Incident coordination
Transport Intelligence	Mobility optimization
Volunteer Intelligence	Task allocation
Executive Analytics	KPI & reporting

AI Architecture

User



Prompt Builder



Context Injection



RAG



Gemini



429 / Failure?



Groq



Function Calling



Response

Data Flow

Sensor/API



Backend Service



Firestore

↓
AI Gateway
↓
Digital Twin
↓
WebSocket
↓
Dashboard

System Workflows

Workflow 1 — Fan Journey

Ticket Purchased
↓
Travel Planning
↓
Arrival
↓
Gate Entry
↓
Seat Navigation
↓
Food & Merchandise
↓

Live Match

↓

Exit

↓

Transport Home

ArenaMind AI assists throughout the journey.

Workflow 2 — AI Crowd Prediction

Ticket Scan

↓

Crowd Engine

↓

Density Analysis

↓

AI Prediction

↓

Risk Assessment

↓

Recommendation

↓

Digital Twin Update

↓

Dashboard Notification

Workflow 3 — Emergency Response

Incident Detected

↓

AI Classification

↓

Severity Assessment

↓

Resource Allocation

↓

Medical / Security Dispatch

↓

Crowd Rerouting

↓

Emergency Broadcast

↓

Incident Resolution

↓

AI Report

Workflow 4 — Fan Assistant

User Query

↓

Intent Detection

↓
Navigation / Search
↓
Google Maps
↓
AI Recommendation
↓
Personalized Response

Example:

"Find vegetarian food."
↓
Nearby restaurants
↓
Queue prediction
↓
Walking route
↓
Recommendation

Workflow 5 — AI Command Center

Operator Query
↓
AI Gateway

↓
RAG
↓
Live Stadium Data
↓
Gemini
↓
Groq (Fallback)
↓
Function Calls
↓
Operational Recommendation

Workflow 6 — Digital Twin

Crowd Data
↓
Weather
↓
Transport
↓
Security
↓
Medical
↓
AI Analysis

↓

3D Stadium Update

↓

User Interaction

Workflow 7 — Executive Analytics

Operational Data

↓

Aggregation

↓

AI Summary

↓

KPI Generation

↓

Executive Dashboard

↓

Export Report

Workflow 8 — AI Report Generation

Incidents

↓

Crowd Metrics

↓
Transport
↓
Energy
↓
AI Summary
↓
Executive Report
↓
PDF / PPT Export

Gemini → Groq Failover

User Request

↓

Gemini

↓

Success?

↓

YES

↓

Response

NO

↓

Groq

↓

Continue Session

No interruption to the user.

Event-Driven Workflow

Live Event

↓

Event Bus

↓

Subscribers

↓

AI Analysis

↓

Recommendation

↓

Dashboard Update

↓

Notification

Digital Twin Synchronization

Live Data

↓

Firestore

↓

Digital Twin Engine

↓

3D Visualization

↓

User Dashboard

Authentication Flow

User

↓

Firebase Auth

↓

JWT

↓

API Gateway

↓

Role Validation

↓

Authorized Module

AI Function Calling

User Prompt

↓

LLM

↓

Tool Selection

↓

Backend Service

↓

Structured Result

↓

Natural Language Response

Deployment Architecture

Users

↓

Vercel

↓

Next.js

↓

FastAPI

↓

Firebase

↓

Gemini / Groq

↓

Google Cloud APIs

Design Principles

- AI-First
- Cloud-Native
- Event-Driven
- Explainable AI
- Human-in-the-Loop
- Modular Services
- Real-Time Synchronization
- Scalable by Design

Technology Summary

Layer	Technology
Frontend	Next.js, React, Tailwind CSS
3D	React Three Fiber, Three.js
Backend	FastAPI
Database	Firestore
Authentication	Firebase Auth
AI	Gemini 2.5 + Groq
Maps	Google Maps Platform
Storage	Firebase Storage
Deployment	Vercel + Firebase

MVP Scope

- ✓ Landing Page
 - ✓ AI Command Center
 - ✓ Digital Twin
 - ✓ Crowd Intelligence
 - ✓ Fan Assistant
 - ✓ Emergency Response
 - ✓ Executive Analytics
 - ✓ Gemini + Groq AI Integration
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Future Scope

- Multi-Stadium Operations
 - Olympics Support
 - Smart City Integration
 - Autonomous AI Agents
 - Apple Vision Pro
 - Drone Integration
 - IoT Sensors
 - CCTV Vision Analytics
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ArenaMind AI

The Generative AI Operating System for FIFA World Cup Stadiums

"Observe • Understand • Predict • Recommend • Orchestrate"

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